

Теорема Джона и объёмные отношения

Homework

1 сентября 2026 года

1. Find the John ellipsoid of the regular simplex in $\mathbb{R}^n$ and check that the factor n in John's theorem cannot be improved for general bodies.
2. Prove $d(\ell_1^n, \ell_2^n) = d(\ell_\infty^n, \ell_2^n) = \sqrt{n}$ for the Banach–Mazur distance, and deduce from John's theorem that $d(X, \ell_2^n) \leq \sqrt{n}$ for every n -dimensional normed space.
3. Define the Löwner ellipsoid and prove that the John ellipsoid of K° is the polar of the Löwner ellipsoid of K .
4. (Kadets–Snobar 1971) Prove that every n -dimensional subspace of a normed space is the range of a projection of norm at most $\sqrt{n}$. (*Hint: John's position and the decomposition of the identity.*)
5. State Ball's reverse isoperimetric inequality (1991) and verify it for the cube: compare the surface area of the unit-volume cube with that of the unit-volume ball and of the regular simplex.
6. Prove $\text{vr}(K) \geq 1$ with equality only for ellipsoids; prove $\text{vr}(K) \leq \sqrt{n}$ for symmetric bodies and $\leq n$ in general; compute vr of the cube and check that the symmetric bound is sharp up to a constant.
7. Prove the Santaló identity $|\mathcal{E}||\mathcal{E}^\circ| = |B_2^n|^2$ for ellipsoids, compute $|K||K^\circ|$ for the cube and the cross-polytope, and state what the Blaschke–Santaló inequality asserts.
8. (ICMC 2022–23) Show that if the distance between opposite edges of a tetrahedron is at least 1, then its volume is at least $1/3$.

9. (IMC) Let $d \geq 2$. Prove that there is $C(d)$ such that for every symmetric convex polytope $K \subseteq \mathbb{R}^d$ and every $\varepsilon \in (0, 1)$ there is a convex polytope L with at most $C(d)\varepsilon^{1-d}$ vertices with $(1 - \varepsilon)K \subseteq L \subseteq K$.
10. Prove John's theorem in full: if B_2^n is the maximal-volume ellipsoid in K , then the contact points admit weights with $\sum c_i u_i \otimes u_i = I$ and $\sum c_i u_i = 0$. (*Hint: otherwise there is a separating hyperplane in the space of symmetric matrices, and the ellipsoid can be inflated slightly.*)